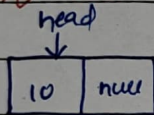


Singly Linked List

Suppose the operations are:-

1. Insert at beginning $\rightarrow 10$
2. Insert at end $\rightarrow 20$
3. Insert at end $\rightarrow 30$
4. Insert at position $\rightarrow 15$ at pos 2
5. display()
6. Search $\rightarrow 20$
7. Reverse
8. Delete At Beginning()
9. Delete At End()

1. Insert at beginning: 10

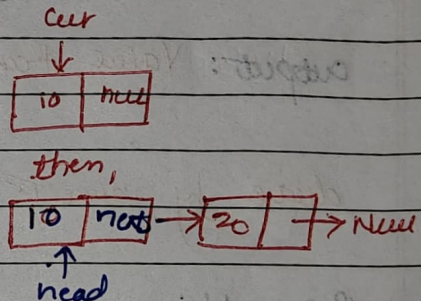


2. Insert at end: 20

head $\neq$ null, so

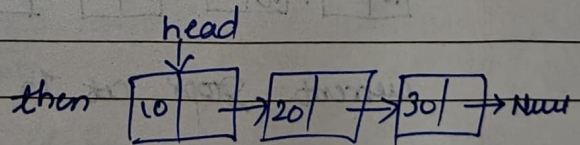
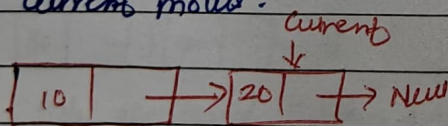
current = head

current.next = new node



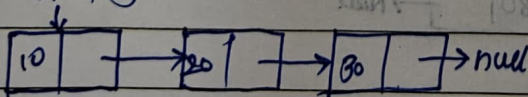
3. Insert at end: 30

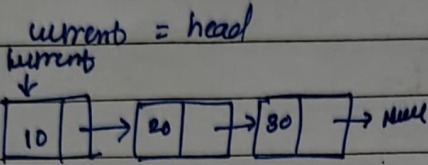
current moved:



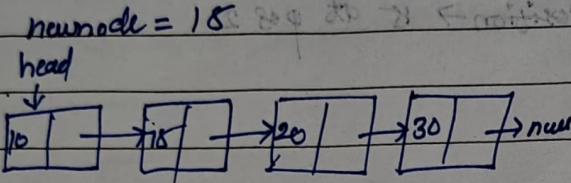
4. Insert ¹⁵ at position 2

initially list:





Now on posⁿ is 2, So loop stop at node 20.
create new node



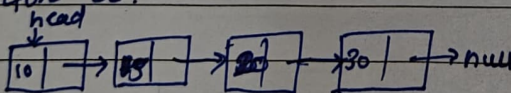
5. Search : 20

current = head
check each node :-
 $10 == 20$ X
 $15 == 20$ X
 $20 == 20$ ✓

output: Value found

6. delete by value 15

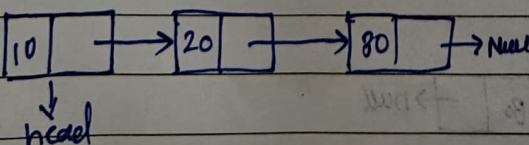
Before LL:



current stops at 10

then,

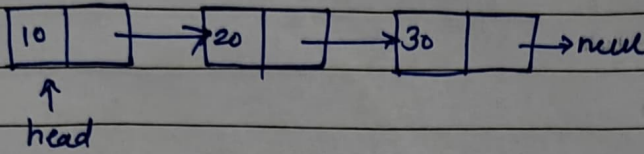
current.next = current.next.next



15 is removed from chain

8. Reverse :-

Before :

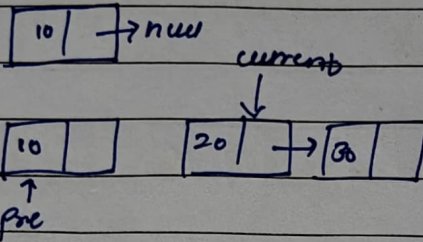


initially :

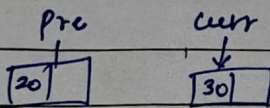
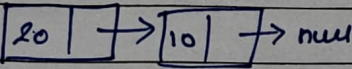
pre = null

curr = 10

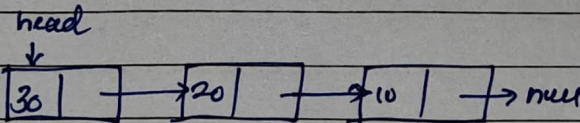
iteration 1:



iteration 2:

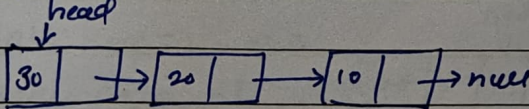


iteration 3:



9. Delete at Beginning :-

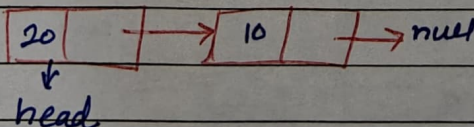
Before :-



int removed = head.data // 30

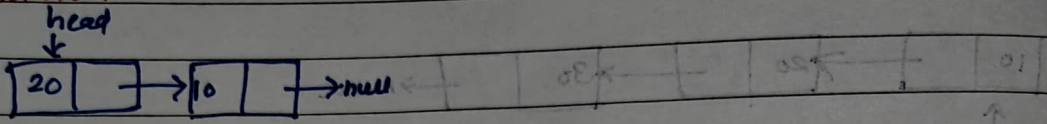
head = head.next

Result :



10. Delete at end :-

Before :

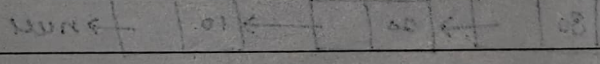
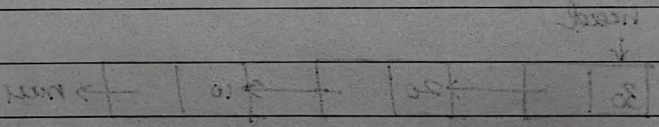
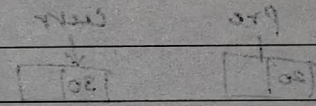
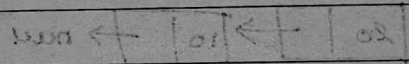
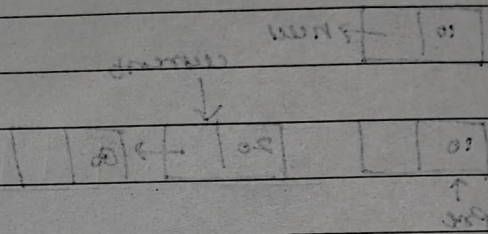
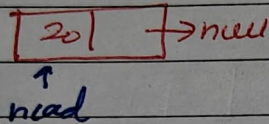


curr moved to 20

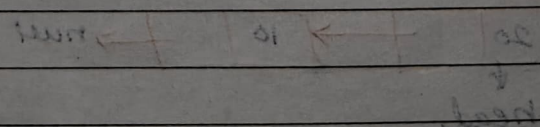
check : $\text{current} \rightarrow \text{next} \rightarrow \text{next} \neq \text{null}$

So loop stop then removed 10.

Result :



1) $\text{current} \rightarrow \text{next} \rightarrow \text{next} \neq \text{null}$
 2) $\text{current} \rightarrow \text{next} = \text{null}$



Doubly Linked List

Suppose the operations are:

1. Insert at Beginning $\rightarrow 10$
2. Insert at end $\rightarrow 20$
3. Insert at position $\rightarrow 30$ at 2
4. Display backward
5. Delete by value $\rightarrow 20$
6. Display Forward

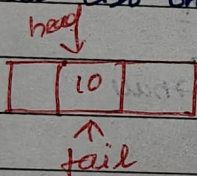
1. Insert at Beginning $\rightarrow 10$

initially:

head = null, tail = null

Create: newnode = [10]

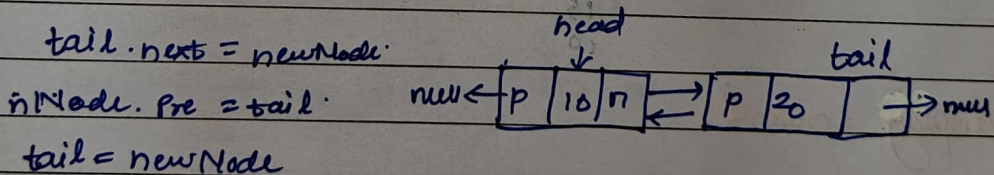
Since List Empty head = tail = newnode



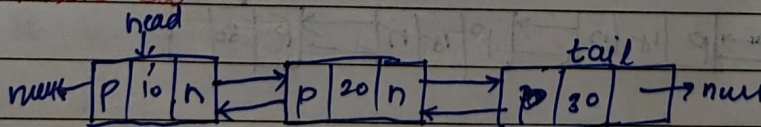
Both head & tail pt to same

2. Insert at End $\rightarrow 20$

Create: newnode = [20]

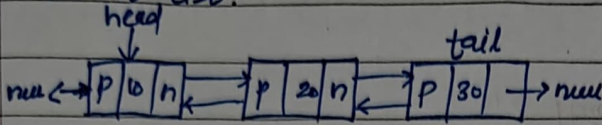


3. Insert at End $\rightarrow 30$



4. Insert at position 2 $\rightarrow 15$

current list:

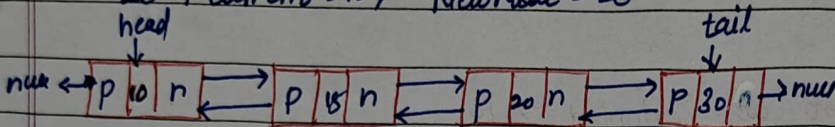


posⁿ = 2

current = head

Create: newNode [15], newNode = current->next

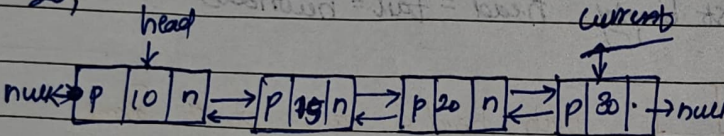
So, current = 10, New Node = 20



5. Display Backward:

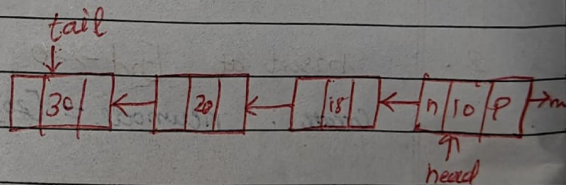
current = tail

So,



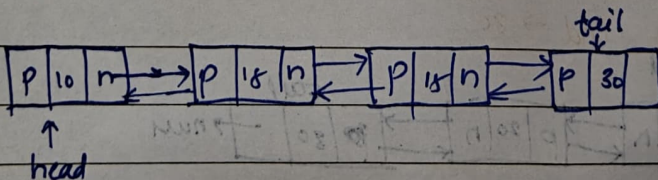
current = current->prev;

o/p:- Backward: 30
20
15
10



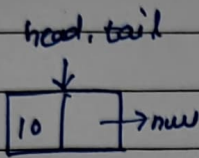
6. display forward:

current = head

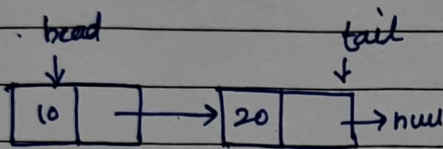


Deque Using Singly Linked List

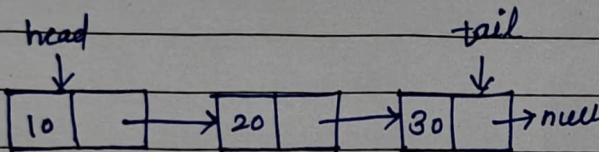
1. insertFront(20)



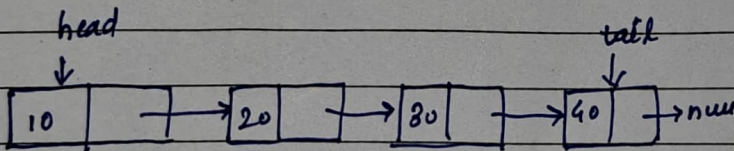
2. insertFront(10)



3. insertRear(30)

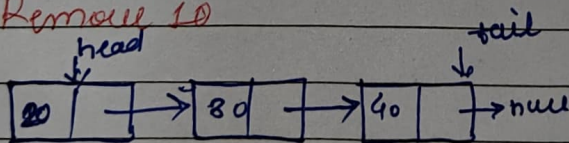


4. insertRear(40)



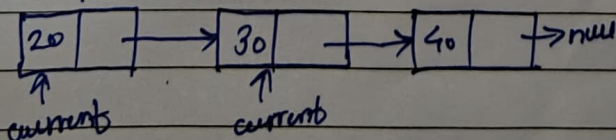
5. deleteFront()

Remove 10



6. deleteRear()

Remove 40, Since this is SLL, we cannot directly go backward from Tail
So we traverse:



Then:

tail = current

tail->next = null

final:

